# A Prototype-Aware Learning and Dual-View Regularization Network for Weakly Supervised Change Detection in VHR Remote Sensing Images

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***Abstract*—The change detection (CD) is a critical task for monitoring the spatiotemporal evolution of the Earth's surface. Recently, due to the advantages of reduced annotation cost and improved labeling efficiency, weakly supervised CD (WSCD) has attracted increasing attention. However, existing WSCD methods encounter several critical challenges, including incomplete activation of class activation maps (CAMs), interference from noisy pseudo-labels during training, and instability in change recognition caused by illumination and environmental variations. To address these issues, we propose a prototype-aware learning and dual-view regularization network (PDRNet) for image-level WSCD. Specifically, to address the issue of incomplete activation caused by the tendency of CAM to focus excessively on locally discriminative regions, PDRNet devises a prototype-aware module (PAM), which captures stable category prototypes and refines CAM quality by reactivating hierarchical features. Furthermore, to mitigate the network's sensitivity to noisy pseudo-labels, a dual-view regularization strategy (DRS) is designed to partition pseudo-labels into clean and noisy regions. Region-specific regularization is subsequently employed to improve the robustness of the model against noisy supervision. Finally, to enhance the capability of identifying changed regions, PDRNet constructs a wavelet-based change enhancement module (WCEM) to decompose bitemporal features into multiple frequency bands. This facilitates the comprehensive utilization of low-frequency structural semantics and high-frequency texture details. Extensive experiments and analyses conducted on three publicly available CD datasets yield the superiority of PDRNet.**

***Index Terms*—Change detection (CD), high-resolution remote sensing image, noisy pseudo-labels, prototype learning, weakly supervised learning.**

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## I. Introduction

CHANGE detection (CD) aims to identify and quantify land cover changes in very high-resolution (VHR) remote sensing images captured at different temporal intervals for the same geographic region [1], [2], [3]. CD technology has found extensive applications across a range of fields, including disaster assessment [4], ecological environment monitoring [5], and urban development [6].

Early CD methods predominantly employ algebraic or statistical approaches [7], such as change vector analysis (CVA) [8], principal component analysis (PCA) [9], and multivariate alteration detection (MAD) [10]. Despite their intuitive principles and ease of implementation, these methods rely on handcrafted features, which makes them susceptible to environmental disturbances and significantly constrains their generalization capability [11]. With the advancement of deep learning, many researchers have employed deep neural networks to automatically extract latent patterns and abstract features from images [12], [13], [14], [15], [16], [17], [18]. Common architectures include convolutional neural networks (CNNs) and attention-based models such as Transformers [19], [20]. Beyond these architectures, wavelet transforms have attracted increasing attention for modeling multiscale spatial–frequency information in CD tasks [21], [22], [23]. By decomposing images into different frequency components, wavelet transforms can enhance the temporal interaction modeling and provide complementary structural priors, thereby improving the robustness of spatiotemporal feature fusion within deep learning frameworks [24], [25]. However, these deep learning-based methods typically demand extensive pixel-level annotated data to achieve strong performance [26]. Unfortunately, pixel-level labeling of remote sensing images is time-consuming and labor intensive [27], [28]. This challenge limits the applicability of methods relying on pixel-level supervision in practical scenarios.

To alleviate the substantial reliance on pixel-level annotations, weakly supervised CD (WSCD) establishes a mapping mechanism from coarse-grained supervision signals to pixel-level predictions [29], [30], [31]. This approach substantially reduces the labor-intensive burden of manual labeling [32]. WSCD typically leverages image-level annotations [28] as supervisory signals. Notably, image-level binary labels have

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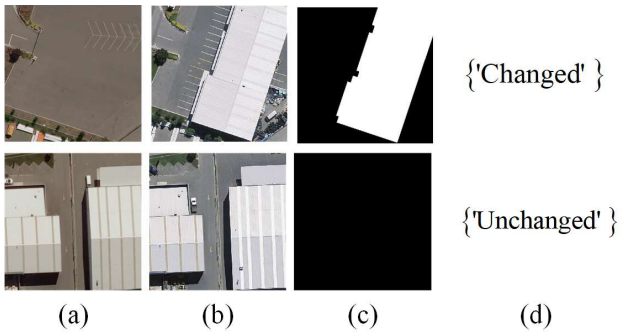



Fig. 1. Comparison of annotations at the pixel level and image level. In the pixel-level labels, white indicates change regions, while black represents unchanged areas. Image-level labels only label bitemporal images as either "changed" or "unchanged." (a) T1 image, (b) T2 image, (c) pixel-level labels, and (d) image-level labels.

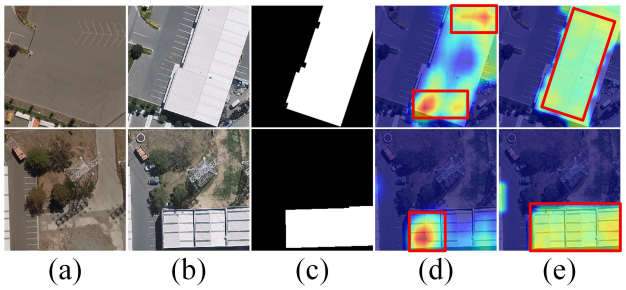



Fig. 2. Initial CAM from the classification network exhibits incomplete activation, with red indicating high response and blue indicating low response. As shown by the red boxes in (d), the initial CAM focuses only on localized changes (e.g., building corners). In contrast, PDRNet yields more complete activations aligned with actual change regions. (a) T1 image, (b) T2 image, (c) pixel-level label, (d) initial CAM, and (e) PDRNet.

garnered significant attention due to their superior labeling efficiency. As shown in Fig. 1, image-level labels only provide whether a change has occurred in the bitemporal image, without directly locating the spatial distribution of the change regions. WSCD aims to derive accurate pixel-level predictions from such coarse-grained image-level supervision, but the semantic gap between image-level semantics and pixel-level localization constitutes a core challenge.

To bridge this gap, class activation map (CAM) [33] is widely adopted as a pivotal technique in weakly supervised learning [34], [35]. By backpropagating semantic responses through a classification network, it enables the extraction of pixel-level localization cues from image-level supervision [36]. Some researchers have incorporated CAM into WSCD to achieve cross-granularity reasoning from image-level labels to pixel-level localization [37]. Wang et al. [38] generated pseudo-labels by integrating multiscale CAMs and fine-tuned EfficientSAM [39] using LoRA technology to improve the performance of CD tasks. Cao et al. [40] utilized single-phase high-resolution images and image-level labels, combining multiscale CAMs with temporal correlation to effectively address pseudo change issues. Wang et al. [41] proposed a WSCD framework that combines the localization capability of CAMs and the zero-shot segmentation ability of the segment anything model (SAM) [42], generating more precise pixel-level pseudo-labels. Dai et al. [43] improved the accuracy of change object boundaries in CAMs by integrating multiscale joint supervision and consistency regularization. Liu et al. [44] employed multiscale self-supervised constraints to bridge the gap between weak and full supervision, while using change masks to maintain the integrity of change regions in CAM.

Although the aforementioned methods have made progress in the field of WSCD, several critical challenges still remain. First, the CAM generation mechanism, which relies on a classification paradigm, is inherently constrained by the optimization properties of the cross-entropy loss function. This constraint compels the network to excessively emphasize highly discriminative local feature regions, thus causing incomplete activation [36], [45]. Specifically, this incomplete activation intuitively manifests as the generated CAM covering only partial regions of the actual changes. As illustrated by the red boxes in Fig. 2(d), the CAM generated by the image-level supervised classification network tends to cover only the corners of buildings or individual structures. Therefore, the pseudo-labels generated from these initial CAMs often fail to accurately represent the true change regions. This compromises the quality of supervision during the training of subsequent CD networks and limits the overall performance improvement.

Furthermore, after generating pseudo-labels, existing WSCD methods typically use them directly as supervision for network training. However, although the quality of the CAM is improved, the generated pseudo-labels still inevitably contain noise (e.g., incorrect change regions and imprecise change boundaries). These WSCD methods fail to account for the adverse impact of noise in pseudo-labels on the detection network's robustness and predictive accuracy. During training, noisy labels introduce incorrect supervisory signals, causing the network to learn suboptimal feature representations and ultimately constraining its performance [46]. Third, in remote sensing CD tasks, factors such as illumination variations can interfere with the extraction of true changes from bitemporal images [47]. As shown in Fig. 2(a) and (b), there is a noticeable color difference between the bitemporal images. These factors may lead the network to misinterpret differences in brightness or color as actual changes, thereby undermining the stability and reliability of the extracted change features. This challenge becomes even more pronounced in complex terrains and environments, where the model struggles to effectively distinguish between changed and unchanged regions [48].

In view of the above issues, a prototype-aware learning and dual-view regularization network (PDRNet) is proposed for weakly supervised CD of VHR remote sensing images. Specifically, to mitigate the tendency of CAM to focus only on the most discriminative change regions, a prototype-aware module (PAM) is designed. The PAM constructs global prototypes as stable representations of category semantics and identifies potential consistent regions through semantic matching, which enhances the completeness of the CAM. Furthermore, to alleviate the impact of noisy pseudo-label on the performance of

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the CD network, a dual-view regularization strategy (DRS) is proposed. This strategy generates dual-view predictions from the same input sample via strong and weak augmentations, and partitions pseudo-labels into clean and noisy regions based on the consistency and confidence. Clean regions guide supervised learning, while noisy regions are regularized with robust loss to reduce noise impact. Finally, to enhance the ability of the network to model temporal change information, a wavelet-based change enhancement module (WCEM) is proposed. By modeling the structural and boundary information of change regions from a frequency perspective, WCEM suppresses the interference caused by illumination and brightness differences, enabling more accurate change perception.

The main contributions can be summarized as follows.

1) A PAM is proposed, which better bridges the semantic gap between image-level semantics and pixel-level predictions. By performing semantic matching between stable class prototypes and fused hierarchical features, it effectively leverages the local details of shallow features and the global semantic information of deep features, enhancing the completeness of CAMs.
2) A DRS is presented, which mitigates the overfitting of the CD network to the noisy pseudo-labels generated from CAMs. It exploits the consistency and confidence between weakly and strongly augmented views to divide pseudo-labels into high-confidence clean regions and uncertain noisy regions. Supervised learning and consistency regularization are then applied to these regions, respectively, to enhance model robustness.
3) A WCEM is designed to fully model temporal correlations, which enhances the sensitivity to change regions. By perceiving global structural changes in the low-frequency component and enhancing texture detail changes in the high-frequency component, it captures more discriminative change features.

The remaining sections are structured as follows. Section II provides a detailed introduction to the proposed PDRNet. Section III presents experimental results on different datasets and analysis. Finally, Section IV draws a conclusion.

## II. Proposed Method

In this section, the overall framework of PDRNet is first introduced. Subsequently, the PAM, DRS, and WCEM are discussed in detail. Finally, the loss function is presented.

### A. Overview

The architecture of the proposed PDRNet pipeline is illustrated in Fig. 3. It consists of two stages: pseudo-label generation and CD network training. In the first stage, a classification network is trained using image-level labels to generate high-quality pseudo-labels. In the second stage, these pseudo-labels are used to train the CD network, producing more accurate pixel-level CD results.

In the pseudo-label generation stage, the network consists of a feature extractor, a classifier, and a PAM. Specifically, for a pair of bitemporal images $\mathbf{I} = \{\mathbf{I}_1, \mathbf{I}_2\} \in \mathbb{R}^{H\times W\times 3}$, where $H$, $W$, and 3 represent the image height, width, and the number of spectral bands, respectively. Its image-level label can be denoted by $y_{\text{img}} \in \{\text{changed}(1), \text{unchanged}(0)\}$, where "1" represents "changed" and "0" indicates "unchanged." The classification network employs a widely used architecture in weakly supervised learning. It utilizes ResNet50 stages 1–4 as the backbone to extract multilevel features $\{\mathbf{F}_1^i, \mathbf{F}_2^i\} \in \mathbb{R}^{H_i\times W_i\times C_i}$, where $i \in \{1, 2, 3, 4\}$ denotes the $i$th stage of the backbone. The hierarchical change feature $\mathbf{F}_d^i \in \mathbb{R}^{H_i\times W_i\times C_i}$ is extracted by the difference operation of the bitemporal features, which can be formulated as follows:

$$\mathbf{F}_d^i = \left|\mathbf{F}_1^i - \mathbf{F}_2^i\right| \tag{1}$$

where $|\cdot|$ denotes the absolute value operation.

The deepest change feature $\mathbf{F}_d^4$ is fed into the classifier for classification and the global average pooling layer is applied to obtain the final classification result $p_{\text{cls}}$. Then, the weight matrix $\mathbf{W}$ of the final classification layer is used to generate the CAM for the change class. This process can be formulated as follows:

$$\mathbf{M}_c = \frac{\text{ReLU}\left(\mathbf{W}^{\text{T}}\mathbf{F}_d^4\right)}{\max\left(\text{ReLU}\left(\mathbf{W}^{\text{T}}\mathbf{F}_d^4\right)\right)} \tag{2}$$

where ReLU denotes the rectified linear unit function, which eliminates negative activation values. max(·) represents min–max normalization, which normalizes the CAM values to the range $[0, 1]$.

Furthermore, the unchanged category activation map $\mathbf{M}_{uc} = 1 - \mathbf{M}_c$ is estimated based on $\mathbf{M}_c$. Subsequently, to generate more refined pseudo-labels, the PAM constructs global prototypes from high-confidence regions in the initial CAM. These prototypes serve as stable representations of category semantics and are used to generate more refined pseudo-labels. By reactivating the fused hierarchical change features with these prototypes, the network enhances feature responses closely associated with the target class. As a result, the spatial completeness of the pseudo-labels is substantially improved.

In the second stage, the pseudo-labels generated in the first stage are used as supervision for the CD network. The detection network is based on a typical U-Net architecture, consisting of an ResNet18 encoder, a decoder, and WCEM. To prevent the network from overfitting to noisy pseudo-labels, DRS is employed during training. By analyzing the consistency and discrepancy between predictions under different data augmentation strategies, pseudo-labels are divided into clean and noisy regions. Different regularization strategies are applied to each region to enhance the robustness of the network. In addition, to enhance the network's ability to capture change regions, a WCEM is employed to facilitate the interaction of bitemporal features. Finally, the decoder progressively upsamples the features layer by layer to generate the final CD result.

### B. Prototype-Aware Module

Due to the optimization characteristics of the cross-entropy loss function, the pseudo-label generation network focuses on the most discriminative local feature regions [49]. As a result, the initial CAMs suffer from severe incomplete activation. Since the prototype essentially reflects the semantic features of the class center, the potential change areas that are not

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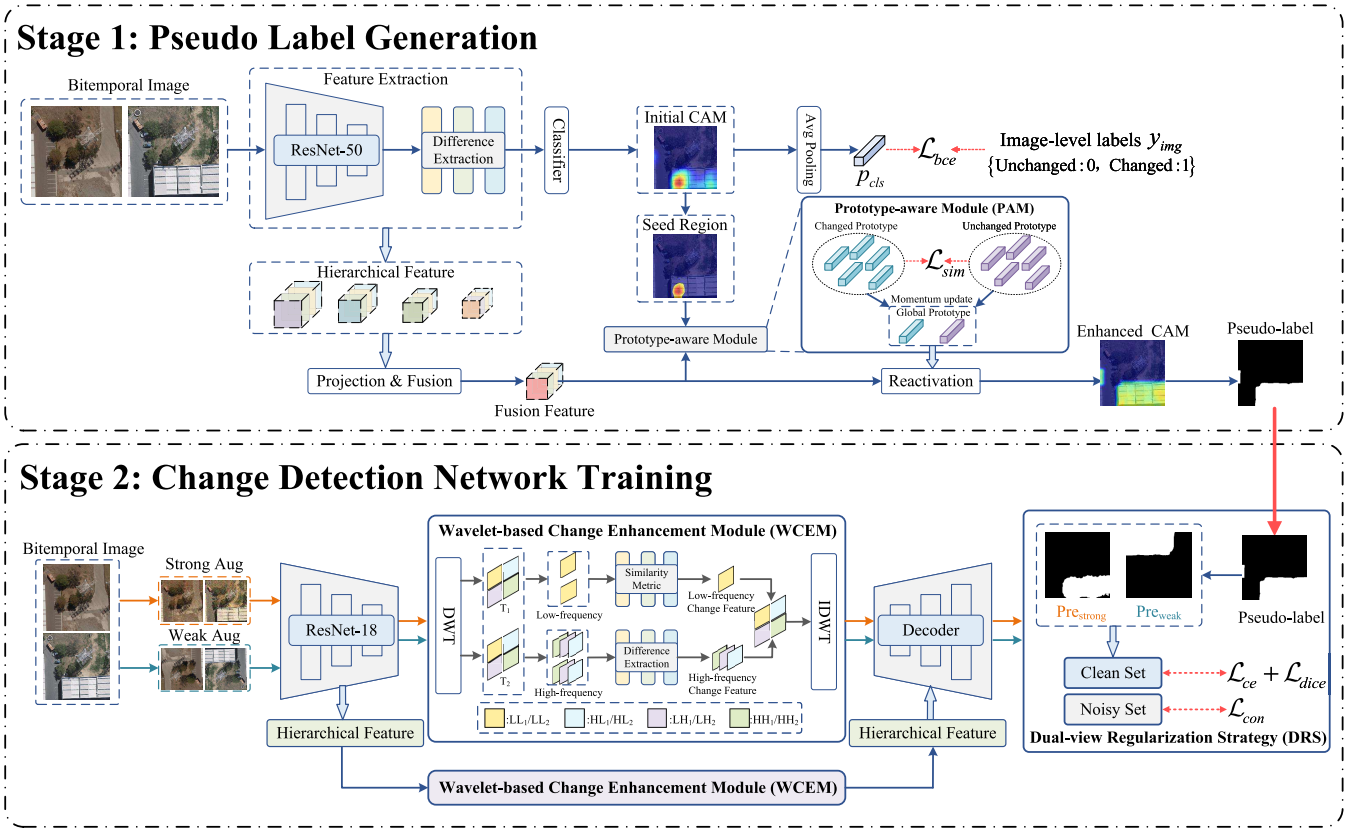



Fig. 3. Overall pipeline of the proposed PDRNet. PDRNet consists of two main stages: pseudo-label generation and CD network training. In stage 1, PAM reactivates fused hierarchical features to refine CAMs, which leads to the generation of high-quality pseudo-labels. In stage 2, the pseudo-labels obtained from stage 1 are used to train the CD network. WCEM is employed to extract change features, while DRS is used to reduce the network overfitting to noise in the pseudo-labels.

activated by the initial CAM can be mined by similarity matching with the feature map. Inspired by this, a PAM is proposed. The PAM aims to construct prototypes using high-confidence regions from the initial CAM. These prototypes are dynamically updated based on the exponential moving average (EMA) to smooth variations and enhance robustness. Ultimately, this module optimizes the CAM by reactivating the fused hierarchical features through prototype-based refinement.

Specifically, a widely used thresholding method is employed to extract seed regions (i.e., high-confidence regions) from $\mathbf{M}_n$, which can be formulated as follows:

$$\hat{\mathbf{M}}_n = \begin{cases} 1, & \text{if } \mathbf{M}_n > \tau \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $\hat{\mathbf{M}}_n \in \mathbb{R}^{h \times w}$ represents the seed region mask for category $n$, $n \in \{c, uc\}$. $\mathbf{M}_n$ denotes the initial CAM, where $\mathbf{M}_c$ and $\mathbf{M}_{uc}$ represent the initial CAMs for the changed and unchanged categories, respectively. $\tau$ denotes the threshold.

Although deep features contain rich semantic information, their low spatial granularity may result in the loss of fine details. Among the hierarchical features extracted by the backbone, shallow features preserve rich boundary and detail information, while deep features provide stronger class discrimination. Therefore, the hierarchical feature fusion is performed to balance semantic information and spatial details, facilitating the generation of more comprehensive and precisely localized prototypes. This process can be formulated as follows:

$$\mathbf{F}_h = \text{Concat}\left( \mathcal{I}\left( \frac{\mathbf{F}_d^i}{\|\mathbf{F}_d^i\|_2 + \epsilon} \right) \middle| \; i = 1, 2, 3, 4 \right) \tag{4}$$

where $\| \cdot \|_2$ denotes the $L2$-norm, $\mathbf{F}_h$ represents the fused hierarchical features, Concat$(\cdot)$ denotes the concatenation operation along the channel dimension, $\mathcal{I}(\cdot)$ refers to bilinear interpolation, and $\epsilon$ is a small constant for numerical stability.

Then, the seed regions are used as guidance to extract class-relevant information from the fused features for prototype computation. Specifically, a weighted aggregation of the fused features is performed based on the seed regions to construct prototype representations that effectively capture class semantics. This can be formulated as follows:

$$\boldsymbol{P}_n = \frac{\sum_{i,j} \hat{\mathbf{M}}_n(x, y)\, \mathbf{F}_h(x, y)}{\sum_{i,j} \hat{\mathbf{M}}_n(x, y)} \tag{5}$$

where $\boldsymbol{P}_n \in \mathbb{R}^D$ represents the prototype of class $n$ for a single instance within a mini-batch. $(x, y)$ denotes the spatial coordinates on the feature map, with $i$ and $j$ iterating over all spatial locations.

For each mini-batch, a similarity loss is introduced between the change prototype $\boldsymbol{P}_c$ and the unchanged prototype $\boldsymbol{P}_{uc}$ to reduce interclass similarity and enhance discriminative capability. Specifically, the cosine similarity between class prototypes is minimized to make them more distinguishable

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in the feature space, thereby optimizing the prototype representations. This process can be formulated as follows:

$$L_{\text{sim}} = -\frac{1}{N}\sum_{i\in N}\log\left(1-\cos\left(\boldsymbol{P}_c^i\cdot\boldsymbol{P}_{uc}^i\right)\right) \tag{6}$$

where $\boldsymbol{P}_c^i$ and $\boldsymbol{P}_{uc}^i$ represent the change and unchanged class prototypes of the $i$th batch, respectively. $N$ represents the size of the mini-batch, and $\cos(\cdot)$ denotes the cosine similarity.

In each mini-batch, the class prototypes for all samples within the batch are first computed and averaged to obtain the batch-level class representation. Subsequently, an EMA is applied for cross-batch updates to obtain a more stable global class prototype. This enables the global prototype to gradually incorporate historical information and undergo dynamic optimization. This process can be formulated as follows:

$$\hat{\boldsymbol{P}}_n^k = \alpha\hat{\boldsymbol{P}}_n^{k-1} + (1-\alpha)\,\bar{\boldsymbol{P}}_n^k \tag{7}$$

where $\hat{\boldsymbol{P}}_n^k$ represents the global prototype of class $n$ at the $k$th iteration, $\bar{\boldsymbol{P}}_n^k$ denotes the averaged prototype of class $n$ within a mini-batch at the $k$th iteration, and $\alpha$ is the momentum coefficient.

After obtaining the global class prototypes, enhanced CAMs are generated by computing the cosine similarity between the fused hierarchical features $\mathbf{F}_h$ and the class prototypes $\hat{\boldsymbol{P}}_n$. The above process can be formulated as follows:

$$\mathbf{S}_n(x,y) = \frac{\hat{\boldsymbol{P}}_n\cdot\mathbf{F}_h(x,y)}{\left\|\hat{\boldsymbol{P}}_n\right\|_2\left\|\mathbf{F}_h(x,y)\right\|_2} \tag{8}$$

where $\mathbf{S}_n$ represents the reactivation map for class $n$.

Finally, for each pixel location, the class with the highest similarity is selected as the pseudo-label for that pixel, which can be formulated as follows:

$$\mathbf{Y}_{\text{pse}} = \begin{cases}\arg\max(\mathbf{S}_n), & \text{if } \max(\mathbf{S}_n) > \beta\\ 0, & \text{otherwise}\end{cases} \tag{9}$$

where $\mathbf{Y}_{\text{pse}}$ represents the pseudo-labels and $\beta$ is the threshold coefficient used to balance the background (unchanged) regions.

### C. Dual-View Regularization Strategy

Although the quality of CAM has been improved, the generated pseudo-labels still inevitably contain erroneous change regions or inaccurate change boundaries. During training, noisy pseudo-labels are prone to introducing incorrect supervisory signals, thereby limiting overall performance. To address this issue, DRS is designed to generate predictions from different perspectives by leveraging strong and weak augmentations to the bitemporal images. Based on these results, the pseudo-labels are classified into clean and noisy regions. Corresponding regularization strategies are applied to different regions, thereby effectively reducing the impact of noisy pseudo-labels and enhancing the robustness of the network.

Specifically, for the input bitemporal image pair $\mathbf{I}$, strong and weak augmentation operations are first applied separately to obtain the strongly augmented sample $\mathbf{I}_s$ and the weakly augmented sample $\mathbf{I}_w$. The strong augmentation involves adding Gaussian noise and performing color transformation, which introduces perturbations to highlight label uncertainty. In contrast, the weak augmentation consists of horizontal and vertical flipping, which maintains the structural integrity of the image to provide more reliable predictions. This complementary design enables the comparison of the two prediction results, thereby dividing pseudo-labels into clean and noisy regions. Subsequently, the two-view samples are fed into the same network to obtain prediction results. Taking the weakly augmented sample $\mathbf{I}_w$ as an example, this process can be formulated as follows:

$$\mathbf{p}_w = f(\mathbf{I}_w;\theta) \tag{10}$$

where $\mathbf{p}_w \in \mathbb{R}^{H\times W\times 2}$ represents the prediction result of the weakly augmented sample, $f(\cdot)$ denotes the CD model, and $\theta$ represents the model parameters. Similarly, the prediction result $\mathbf{p}_s \in \mathbb{R}^{H\times W\times 2}$ for the strongly augmented sample is also obtained via (10).

The reliability of pseudo-labels can be effectively distinguished based on the differences and consistencies between predictions from strongly and weakly augmented views. If predictions from both views are consistent with the pseudo-label at a pixel and their confidences exceed a threshold $\sigma$, the pixel is regarded as a high-confidence clean sample. Otherwise, it is assigned to the noisy set. This process can be formulated as follows:

$$\begin{aligned}\mathcal{C} = &\left\{(i,j)\,\middle|\,\mathbf{p}_w\left(\mathbf{Y}_{\text{pse}}^{(i,j)};(i,j)\right) > \sigma\right\}\\ &\cap\left\{(i,j)\,\middle|\,\mathbf{p}_s\left(\mathbf{Y}_{\text{pse}}^{(i,j)};(i,j)\right) > \sigma\right\}\end{aligned} \tag{11}$$

$$\mathcal{N} = 1-\mathcal{C} \tag{12}$$

where $\mathcal{C}\in\mathbb{R}^{H\times W}$ and $\mathcal{N}\in\mathbb{R}^{H\times W}$ denote the masks for the clean and noisy regions, respectively, $(i,j)$ denotes the spatial location of a pixel, $\mathbf{Y}_{\text{pse}}^{(i,j)}$ is the pseudo-label at location $(i,j)$, $\mathbf{p}_w(\mathbf{Y}_{\text{pse}}^{(i,j)};(i,j))$ and $\mathbf{p}_s(\mathbf{Y}_{\text{pse}}^{(i,j)};(i,j))$ denote the confidence scores of both views on the pseudo-label category.

After obtaining $\mathcal{C}$ and $\mathcal{N}$, two distinct supervision strategies are employed for training accordingly. For the clean regions $\mathcal{C}$, we apply the cross-entropy loss to supervise the training. The aforementioned process can be formulated as follows:

$$\mathcal{L}_{\text{ce}} = -\frac{1}{|2\mathcal{C}|}\sum_{(i,j)\in\mathcal{C}}\left[\mathbf{Y}_{\text{pse}}^{(i,j)}\log\mathbf{p}_w^{(i,j)} + \mathbf{Y}_{\text{pse}}^{(i,j)}\log\mathbf{p}_s^{(i,j)}\right]. \tag{13}$$

Moreover, in CD tasks, change regions typically represent only a small proportion of the total pixels, resulting in significant class imbalance. To address this issue, the dice loss is additionally introduced to enhance the model's sensitivity to change categories. Let $\mathcal{L}_w$ and $\mathcal{L}_s$ be the dice losses for the weakly and strongly augmented samples, respectively. Taking the weakly augmented sample as an example, its dice loss can be formulated as follows:

$$\mathcal{L}_w = 1-\frac{2\sum_{(i,j)\in\mathcal{C}}\mathbf{Y}_{\text{pse}}^{(i,j)}\mathbf{p}_w^{(i,j)}}{\sum_{(i,j)\in\mathcal{C}}\mathbf{Y}_{\text{pse}}^{(i,j)} + \sum_{(i,j)\in\mathcal{C}}\mathbf{p}_w^{(i,j)}}. \tag{14}$$

Therefore, the final Dice loss can be formulated as follows:

$$\mathcal{L}_{\text{dice}} = \frac{1}{2}\left(\mathcal{L}_w + \mathcal{L}_s\right). \tag{15}$$

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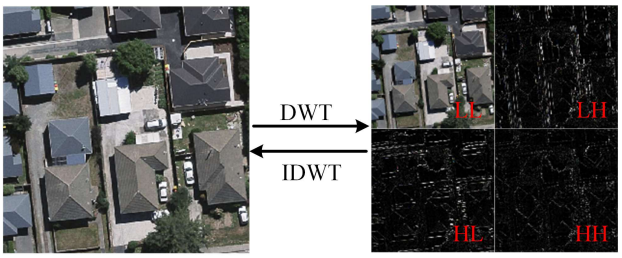



Fig. 4. After performing DWT, the original image is decomposed into four subbands with different frequency components. LL represents low-frequency information, while LH, HL, and HH correspond to high-frequency information.

For the noisy regions $\mathcal{N}$, inspired by the co-training framework, consistency regularization is applied to enforce constraints. This leverages the consistency information between strongly and weakly augmented views, encouraging the model to produce stable predictions in low-confidence regions. The aforementioned process can be formulated as follows:

$$\mathcal{L}_{\text{con}} = -\frac{1}{|\mathcal{N}|} \sum_{(i,j)\in\mathcal{N}} \left[ \mathbf{Y}_w^{(i,j)} \log \mathbf{p}_s^{(i,j)} + \mathbf{Y}_s^{(i,j)} \log \mathbf{p}_w^{(i,j)} \right] \tag{16}$$

where $\mathbf{Y}_w$, $\mathbf{Y}_s \in \mathbb{R}^{H\times W}$ represent the prediction results of the weakly and strongly augmented samples, respectively. By enforcing this constraint, the model is prevented from overfitting noisy pseudo-labels, ultimately improving its generalization ability.

### D. Wavelet-Based Change Enhancement Module

In CD, change features are typically obtained by subtracting or concatenating bitemporal features followed by convolution operations. However, directly processing features is susceptible to variations in illumination and brightness, which can compromise the accuracy of change extraction. To address this challenge, we integrate WCEM to model the low-frequency structural information and high-frequency texture details of change regions, enabling accurate change perception.

Specifically, WCEM employs the discrete wavelet transform (DWT) [50] to decouple bitemporal features into different frequency components [51].

As shown in Fig. 4, after applying DWT to the remote sensing image, it is decomposed into four different subbands. In the low-frequency components (LL), it captures the global structural information of the image, allowing the network to focus on large-scale content such as object shapes and stable background patterns [52], [53]. Meanwhile, in the high-frequency components (LH, HL, and HH), it emphasizes edges, textures, and other fine details, thereby strengthening the representation of local features [54].

In WCEM, the simplest and computationally efficient Haar wavelet is adopted. Its filters consists of a low-pass filter $L = 1/(2)^{1/2}[1, 1]$ and a high-pass filter $H = 1/(2)^{1/2}[1, -1]$. These filters are used to construct four kernels, which operate with a stride of 2 to decompose the input into four subbands: one low-frequency component (LL) and three high-frequency components in the horizontal (LH), vertical (HL), and diagonal (HH) directions. Specifically, the module first applies DWT to the bitemporal features $\{\mathbf{F}_1, \mathbf{F}_2\} \in \mathbb{R}^{H_i\times W_i\times C_i}$ extracted by the encoder, decomposing them into four subbands: $\{\mathbf{F}_1^{\text{LL}}, \mathbf{F}_1^{\text{LH}}, \mathbf{F}_1^{\text{HL}}, \mathbf{F}_1^{\text{HH}}\}$ and $\{\mathbf{F}_2^{\text{LL}}, \mathbf{F}_2^{\text{LH}}, \mathbf{F}_2^{\text{HL}}, \mathbf{F}_2^{\text{HH}}\}$. Their sizes are $H_i/2 \times W_i/2$.




Fig. 5. For the red star-shaped region in the upper right, it corresponds to a tree in the T1 image. Its similarity with surrounding regions such as the cement ground (orange star) and buildings (yellow star) is relatively low, while its similarity with other trees (blue star) is high. In the T2 image, after the red star region is transformed into a building, its similarity relationships with the other star-shaped regions also change accordingly. It exhibits high similarity with the buildings (yellow star) and low similarity with the trees (blue star).

For the low-frequency components $\mathbf{F}_1^{\text{LL}}$ and $\mathbf{F}_2^{\text{LL}}$, they contain most of the image's energy and preserve rich global semantic information. Directly comparing the difference between low-frequency features $\mathbf{F}_1^{\text{LL}}$ and $\mathbf{F}_2^{\text{LL}}$ may be prone to interference from variations in illumination and brightness. To address this issue, WCEM utilizes spatial similarity relationships to improve robustness. As shown in Fig. 5, when the red star-shaped region in the upper right experiences a change (from grassland to a building), its relationship with surrounding objects, such as similarity, also changes accordingly. The spatial relationships among ground objects within the same temporal phase are relatively less influenced by illumination or brightness variations. Therefore, capturing changes in these relationships across different temporal phases enables the extraction of more reliable change information [55].

Specifically, the WCEM first computes the similarity relationships between each pixel and all surrounding pixels within each temporal feature map. This pixelwise similarity aims to capture the structural context at each temporal. This process can be formulated as follows:

$$\text{Sim}_t = \frac{\mathbf{F}_t^{\text{LL}} \cdot \left(\mathbf{F}_t^{\text{LL}}\right)^{\text{T}}}{\left\|\mathbf{F}_t^{\text{LL}}\right\|_2^2} \tag{17}$$

where $\text{Sim}_t \in \mathbb{R}^{(H_iW_i/4)\times(H_iW_i/4)}$ represents a compact representation of structural relationships within the $t$th temporal phases.

Subsequently, by comparing $\text{Sim}_1$ and $\text{Sim}_2$, potential structural change information can be revealed. The aforementioned process can be formulated as follows:

$$\mathbf{D} = \frac{\text{Sim}_1 \cdot \text{Sim}_2}{\|\text{Sim}_1\|_2 \|\text{Sim}_2\|_2} \tag{18}$$

where $\mathbf{D} \in \mathbb{R}^{(H_i/2)\times(W_i/2)}$ represents the change information between the bitemporal images.

In $\mathbf{D}$, the high-confidence regions identified by the network tend to approach 1 and $-1$. Therefore, $\mathbf{D}$ can be used to highlight accurate change information while reducing the

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interference caused by illumination differences and noise. Finally, the change information after being adjusted by $\mathbf{D}$ can be formulated as follows:

$$\mathbf{F}_{\text{dif}}^{\text{LL}} = \left|\mathbf{F}_1^{\text{LL}} - \mathbf{F}_2^{\text{LL}}\right| \odot |\mathbf{D}| \tag{19}$$

where $\mathbf{F}_{\text{dif}}^{\text{LL}}$ represents the change features in the low-frequency component and $\odot$ denotes the Hadamard product operation.

For the high-frequency components, they primarily capture texture and edge details in the image. To comprehensively capture changes in image details, WCEM concatenates the high-frequency subbands from both temporal instances to form a complete high-frequency feature map. Then, the change features are obtained through the difference calculation. This process can be formulated as follows:

$$\mathbf{F}_t^H = \text{Concat}\left(\mathbf{F}_t^{\text{LH}}, \mathbf{F}_t^{\text{HL}}, \mathbf{F}_t^{\text{HH}}\right) \tag{20}$$

$$\mathbf{F}_{\text{dif}}^H = \text{Conv}_{3\times3}\left(\left|\mathbf{F}_1^H - \mathbf{F}_2^H\right|\right) \tag{21}$$

where $\mathbf{F}_t^H$ represents the concatenated high-frequency features and $\mathbf{F}_{\text{dif}}^H$ denotes the change features in the high-frequency component. Concat$(\cdot)$ denotes the concatenation operation along the channel dimension, while $\text{Conv}_{3\times3}(\cdot)$ refers to the convolutional operation with a kernel size of 3.

Finally, $\mathbf{F}_{\text{dif}}^H$ is split back into three subbands: LH, HL, and HH, resulting in $\mathbf{F}_{\text{dif}}^{\text{LH}}$, $\mathbf{F}_{\text{dif}}^{\text{HL}}$, and $\mathbf{F}_{\text{dif}}^{\text{HH}}$. The final change features are obtained through inverse DWT (IDWT).

### E. Loss Function

The overall network architecture of PDRNet consists of two stages: pseudo-label generation and CD network training.

In the pseudo-label generation stage, binary cross-entropy (BCE) loss and prototype similarity loss are employed as the comprehensive loss function

$$\mathcal{L} = \mathcal{L}_{\text{bce}} + \lambda_1 \mathcal{L}_{\text{sim}} \tag{22}$$

$$\mathcal{L}_{\text{bce}} = \frac{1}{N}\sum_{n=1}^{N}\left[-y_{\text{img}}^n \log\left(p_{\text{cls}}^n\right) - \left(1 - y_{\text{img}}^n\right)\log\left(1 - p_{\text{cls}}^n\right)\right] \tag{23}$$

where $\lambda_1$ is a hyperparameter used to balance different loss terms, $N$ represents the batch size, $y_{\text{img}}^n$ is the image-level label of the $n$th sample pair, and $p_{\text{cls}}^n$ denotes its corresponding classification prediction.

In the CD network training stage, the overall loss function is a weighted combination of cross-entropy loss $\mathcal{L}_{\text{ce}}$, dice loss $\mathcal{L}_{\text{dice}}$, and consistency loss $\mathcal{L}_{\text{con}}$

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{dice}} + \lambda_2 \mathcal{L}_{\text{con}} \tag{24}$$

where $\lambda_2$ is a hyperparameter used to balance different loss terms, controlling the contribution weight of $\mathcal{L}_{\text{con}}$ in the overall loss function. Only a single hyperparameter $\lambda_2$ is introduced to control the consistency loss, which reduces the burden of hyperparameter tuning. The cross-entropy loss and the dice loss are assigned equal weights, as their complementary roles in CD tasks have been widely adopted [47], [56], [57].

## III. Experiments

In this section, three publicly available datasets used in the experiments are first introduced, followed by a description of the relevant experimental settings. Subsequently, ablation studies are conducted on the three modules of PDRNet. Then, experimental results of various methods on different datasets are presented. Finally, parameter and complexity analyses of PDRNet are provided.

### A. Dataset

Experiments are conducted on three widely adopted datasets: LEVIR-CD [58], WHU-CD [59], and GZ [60]. The detailed information for each dataset is as follows.

*1) LEVIR-CD:* This dataset contains a total of 637 pairs of high-resolution remote sensing images with a resolution of 0.5 m, each sized $1024 \times 1024$. It focuses on building changes and covers various types of structures, including villas, apartments, and garages. To facilitate training, the images were cropped into nonoverlapping patches of size $256 \times 256$ and split into training, validation, and testing sets in a 7:1:2 ratio. The final dataset consists of 7120, 1024, and 2048 pairs of images for training, validation, and testing, respectively.

*2) WHU-CD:* The resolution of this dataset is 0.075 m, with a size of $32\,507 \times 15\,354$ pixels, primarily capturing changes in buildings. It records building changes in Christchurch, New Zealand, from 2012 to 2016. To facilitate training, the original data was cropped into nonoverlapping patches of size $256 \times 256$ and randomly divided. Ultimately, the training, validation, and testing sets include 5950, 742, and 742 pairs of images, respectively.

*3) GZ:* The dataset records suburban building changes in Guangzhou, China, from 2006 to 2019. It was collected from Google Earth with a spatial resolution of 0.55 m and an image size ranging from $1006 \times 1168$ to $4936 \times 5224$ pixels. The original data were cropped into nonoverlapping patches of size $256 \times 256$ and randomly divided. Ultimately, the training, validation, and test sets consist of 2504, 313, and 313 pairs of images, respectively.

### B. Experimental Details

All experiments are conducted on the Pytorch platform using a single NVIDIA GeForce RTX 3090 24-GB GPU. The batch size is 16. For the pseudo-label generation network, the model is trained for 8 epochs using the SGD optimizer with a momentum of 0.9 and a weight decay of 1e−4. The learning rates for the backbone layers and newly added layers are set to 0.1 and 1, respectively. A poly learning scheduler decayed with a power of 0.9 for the learning rate. The momentum coefficient $\alpha$ is empirically set to 0.9 [37]. $\tau$ is set to 0.7. $\beta$ is empirically set to 0.45 [31]. For the CD network, AdamW is used for training over 50 epochs with a weight decay of 0.01 and an initial learning rate of 0.001. The learning rate is adjusted using a cosine annealing strategy, with a minimum learning rate set to 0.0001. The confidence threshold $\sigma$ is set to 0.9, and the loss weight factor $\lambda$ is set to 0.004.

Following established evaluation practices, five key metrics are employed to quantitatively assess the performance of the

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TABLE I
ABLATION STUDY RESULTS OF PAM ON THREE DIFFERENT DATASETS

| Methods | LEVIR-CD | | WHU-CD | | GZ | |
|---|---|---|---|---|---|---|
| | F1 | IoU | F1 | IoU | F1 | IoU |
| Baseline | 59.14 | 41.99 | 51.53 | 34.72 | 55.11 | 38.03 |
| +PAM | **64.27** | **47.36** | **73.87** | **58.57** | **64.10** | **47.17** |

method: overall accuracy (OA), intersection over union (IoU), $F1$ score ($F1$), precision (Pre.), and recall (Rec.) [61]. The aforementioned five metrics can be formulated as follows:

$$\begin{aligned} \text{Pre} &= \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{Rec} = \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{F1} &= \frac{2\times\text{Pre}\times\text{Rec}}{\text{Pre}+\text{Rec}}, \quad \text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \end{aligned} \tag{25}$$

where TP, TN, FP, and FN denote the number of true positive, true negative, false positive, and false negative, respectively. Among these evaluation metrics, OA reflects the proportion of correctly classified pixels. Precision measures the correctness of predicted change pixels, while Recall indicates the proportion of actual changes correctly detected. IoU evaluates the overlap between predicted and ground truth change regions. $F1$, the harmonic mean of precision and recall, offers a balanced measure of model performance. Compared with other metrics, $F1$ and IoU provide more comprehensive evaluations by jointly considering both accuracy and completeness of CD.

### C. Ablation Studies

In this section, comprehensive ablation studies are performed on the three datasets to rigorously assess the individual contribution of each component within PDRNet and to validate the effectiveness of the overall framework. Specifically, in the pseudo-label generation stage, the baseline network only consists of ResNet50 and a classifier. For the CD network, different network variants are designed by removing one of the components while keeping others consistent, in order to verify the effectiveness of each component.

*1) Effectiveness of PAM:* To evaluate the effectiveness of PAM, pseudo-labels are generated using (9) based on the CAMs produced by the baseline and the baseline enhanced with PAM, respectively. As shown in Table I, the incorporation of PAM substantially enhances pseudo label quality on three datasets, yielding improvements of 5.13%, 22.34%, and 9.01% in $F1$ score, and 5.37%, 23.85%, and 9.14% in IoU, respectively.

Fig. 6 provides a more intuitive illustration of the comparative results from the PAM ablation study across the three datasets. As highlighted by the red boxes in Fig. 6(d), on the WHU-CD and GZ datasets, the CAMs generated by the baseline are limited in capturing large-scale building changes, as they typically focus only on structures or small parts of individual buildings. When dealing with densely distributed small building changes in the LEVIR-CD, the CAMs produced by the baseline also exhibit limitations in completeness and coverage, failing to capture the full extent of the change targets. As illustrated in the red box of Fig. 6(e), following the reactivation of hierarchical features via stable class prototypes within PAM, the resulting CAMs exhibit a more complete representation of change targets along with improved boundary delineation. These findings further validate the effectiveness of the proposed PAM module.

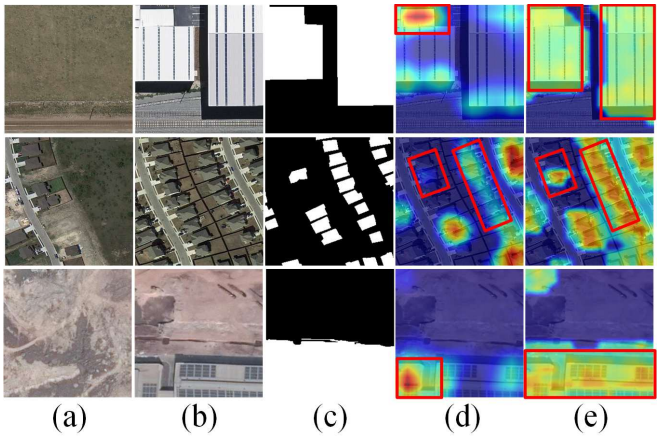



Fig. 6. Visual comparison between the CAMs produced by the baseline and those refined through the integration of PAM. The first, second, and third rows correspond to samples from the WHU-CD, LEVIR-CD, and GZ datasets, respectively. (a) T1 image, (b) T2 image, (c) ground truth, (d) baseline, and (e) baseline+PAM.

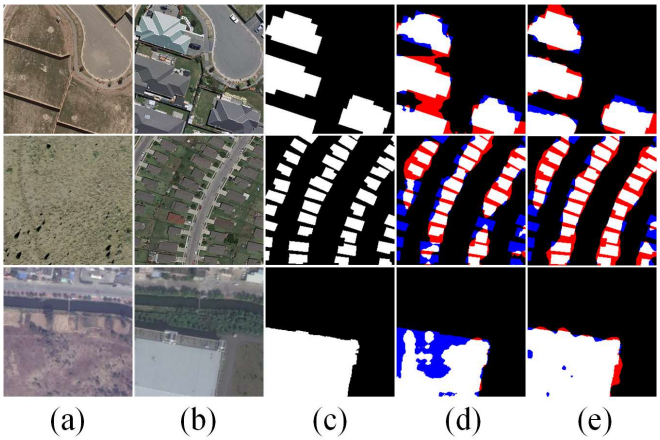



Fig. 7. Comparison of prediction results before and after incorporating DRS on the three datasets. In the prediction results, white pixels represent true positives, black pixels represent true negatives, red pixels represent false positives, and blue pixels represent false negatives. (a) T1 image, (b) T2 image, (c) ground truth, (d) without DRS, and (e) PDRNet.

*2) Effectiveness of DRS:* DRS mitigates the impact of pseudo-label noise on the performance of the CD network through a regularization strategy. As shown in the first and second rows of Table II, the integration of DRS leads to improvements in $F1$ score by 0.64%, 1.73%, and 1.11%, and in IoU by 0.71%, 2.22%, and 1.24% on the three datasets, respectively.

As illustrated in Fig. 7, the integration of DRS leads to a reduction in both false positives and false negatives, particularly in areas such as roads and concrete surfaces that share similar characteristics with buildings. Moreover, the results in the second and third rows indicate that incorporating DRS enables the detection of more complete change regions. This

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TABLE II
ABLATION STUDY RESULTS OF DRS AND WCEM ON THREE DIFFERENT DATASETS

| DRS | WCEM | LEVIR-CD | | | | WHU-CD | | | | GZ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | F1 | IoU | Pre. | Rec. | F1 | IoU | Pre. | Rec. | F1 | IoU | Pre. | Rec. |
| × | × | 65.16 | 48.33 | 54.80 | 60.37 | 74.17 | 58.94 | 64.97 | **86.39** | 65.26 | 48.43 | 57.44 | 75.55 |
| ✓ | × | 65.80 | 49.04 | 55.26 | **81.32** | 75.90 | 61.16 | 70.21 | 82.59 | 66.37 | 49.67 | 56.70 | **80.00** |
| × | ✓ | 65.81 | 49.05 | **56.51** | 78.80 | 75.74 | 60.96 | 71.69 | 80.28 | 66.13 | 49.40 | **60.57** | 72.81 |
| ✓ | ✓ | **66.42** | **49.71** | 56.14 | 81.29 | **77.42** | **63.15** | **73.41** | 81.89 | **67.30** | **50.72** | 59.85 | 76.86 |

improvement can be attributed to the effective regularization strategy of DRS, which mitigates the impact of label noise to a certain extent.

*3) Effectiveness of WCEM:* To assess the effectiveness of WCEM, comprehensive ablation studies are conducted. Specifically, in the absence of WCEM, all instances of the module are replaced with a conventional concatenation operation commonly employed in CD networks, consisting of a standard $3 \times 3$ convolution followed by a $1 \times 1$ convolution. As reported in Table II, the incorporation of WCEM results in performance improvements on three datasets, with $F1$ scores increasing by 0.65%, 1.57%, and 0.87%, and IoU scores increasing by 0.72%, 2.02%, and 0.97%, respectively.

Furthermore, additional experiments are conducted to investigate the impact of different feature interaction approaches. Several fundamental approaches commonly employed in CD are evaluated, including bitemporal feature subtraction, concatenation, and absolute difference. As presented in Table III, WCEM consistently outperforms all baseline interaction methods across the three datasets. This can be attributed to the ability of WCEM to fully exploit low-frequency structural cues and high-frequency textural details, thereby enhancing the capability of the network to accurately and comprehensively delineate change regions.

### D. Comparison With State-of-the-Art Methods

To assess the comprehensive performance of PDRNet, comparative experiments are performed against six state-of-the-art methods on the same experimental platform. These methods include FCD-GAN [28], CS-WSCDNet [41], MS-Former [62], TransWCD [63], TransWCD-SA [63], and ACWCD [64].

Among these methods, FCD-GAN is a fully convolutional CD framework based on a generative adversarial network (GAN), which achieves WSCD through adversarial training between the segmentor and the discriminator. CS-WSCD integrates the localization capability of CAM with the robust segmentation ability of the SAM, refining the quality of pseudo-label generation. MS-Former focuses on representing change information using patch-level annotations while also supporting image-level WSCD. TransWCD is a transformer-based WSCD framework. TransWCD-SA extends TransWCD by incorporating a scene-adaptive predictor, further augmenting its detection performance. ACWCD is an end-to-end encoder–decoder WSCD framework based on image-level labels, leveraging CAM and Vision Transformers (ViTs) [65] to generate pseudo-labels and perform pixel-level supervision.

TABLE III
ABLATION EXPERIMENTS OF FEATURE INTERACTION APPROACH

| Methods | LEVIR-CD | | WHU-CD | | GZ | |
|---|---|---|---|---|---|---|
| | F1 | IoU | F1 | IoU | F1 | IoU |
| $\lvert\mathbf{F_1} - \mathbf{F_2}\rvert$ | 65.68 | 48.89 | 76.12 | 61.45 | 65.81 | 49.05 |
| $\mathbf{F_1} - \mathbf{F_2}$ | 65.88 | 49.12 | 75.79 | 61.02 | 66.01 | 49.27 |
| Concat | 65.80 | 49.04 | 75.90 | 61.16 | 66.37 | 49.67 |
| WCEM | **66.42** | **49.71** | **77.42** | **63.15** | **67.30** | **50.72** |

All the aforementioned methods are evaluated using the parameter settings specified in their respective studies.

*1) Quantitative Evaluation:* The quantitative results of all methods on the three datasets are presented in Table IV. Compared with the second-best method, PDRNet achieves $F1$ improvements of 5.95%, 0.84%, and 0.82% on LEVIR-CD, WHU-CD, and GZ, respectively. In terms of IoU, it achieves improvements of 6.37%, 1.1%, and 0.58% on LEVIR-CD, WHU-CD, and GZ, respectively.

Among all methods, ACWCD achieves the second-best detection performance on WHU-CD and GZ. This is attributed to its use of attention refinement and prior constraint mechanisms to enhance the quality of pseudo-labels, enabling more accurate pixel-level change localization. However, the use of change attention derived from Transformers to refine CAM tends to produce smoother results, which limits detection performance in densely changed scenarios such as LEVIR-CD. CS-WSCDNet achieves the highest recall on the three datasets, but its precision is relatively low. This is because although SAM possesses strong zero-shot segmentation capabilities, it may overextend into adjacent unchanged regions when processing the coarse localization information provided by CAM. MS-Former achieves high accuracy but exhibits a low recall rate, indicating its limited ability to comprehensively identify change regions. This because MS-Former leverages transformers to recover fine-grained change information from bitemporal image patches and enhances features using prototypes, leading the network to focus more on high-confidence changes within small regions.

In contrast, the proposed PDRNet is specifically designed to emphasize the reliability and structural consistency of change region predictions, rather than merely pursuing higher recall. For example, the WCEM captures global structural information through low-frequency components and strengthens reliable boundaries through high-frequency components, which improves precision but may suppress some uncertain changes, slightly reducing recall. Similarly, the DRS mitigates

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TABLE IV

PERFORMANCE COMPARISON OF DIFFERENT CD METHODS ON LEVIR-CD, WHU-CD, AND GZ DATASETS, RESPECTIVELY. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND THE SECOND BEST RESULTS ARE BLUE. ALL RESULTS OF THE THREE EVALUATION METRICS ARE DESCRIBED AS PERCENTAGES (%)

| Method | LEVIR-CD | | | | | WHU-CD | | | | | GZ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | OA | F1 | IoU | Pre. | Rec. | OA | F1 | IoU | Pre. | Rec. | OA | F1 | IoU | Pre. | Rec. |
| FCD-GAN [28] | 63.52 | 24.96 | 14.26 | 16.19 | 54.37 | 78.15 | 51.44 | 34.63 | 41.40 | 67.92 | 77.90 | 50.33 | 33.63 | 60.64 | 43.02 |
| CS-WSCDNet [41] | 88.77 | 45.44 | 29.40 | 30.19 | 91.84 | 96.56 | 68.63 | 52.25 | 57.56 | 84.98 | 93.13 | 55.52 | 38.43 | 71.69 | 45.30 |
| MS-Former [62] | 95.94 | 60.13 | 42.99 | 60.19 | 60.07 | 97.66 | 66.39 | 53.13 | 82.26 | 59.81 | 91.74 | 53.91 | 36.90 | 57.13 | 51.03 |
| TransWCD [63] | 94.70 | 52.77 | 35.85 | 48.33 | 58.12 | 96.07 | 61.77 | 44.69 | 54.32 | 71.59 | 90.06 | 47.23 | 30.91 | 52.80 | 47.22 |
| TransWCD-SA [63] | 94.84 | 60.47 | 43.34 | 49.60 | 77.43 | 97.04 | 69.20 | 52.90 | 64.21 | 75.03 | 91.06 | 61.87 | 44.79 | 51.86 | 76.65 |
| ACWCD [64] | 94.15 | 59.37 | 42.21 | 45.95 | 83.85 | 97.73 | 76.58 | 62.05 | 70.73 | 83.50 | 92.60 | 66.90 | 50.26 | 57.98 | 79.07 |
| **PDRNet** | 95.81 | 66.42 | 49.71 | 56.14 | 81.29 | 97.88 | 77.42 | 63.15 | 73.41 | 81.89 | 92.93 | 67.30 | 50.72 | 59.85 | 76.86 |

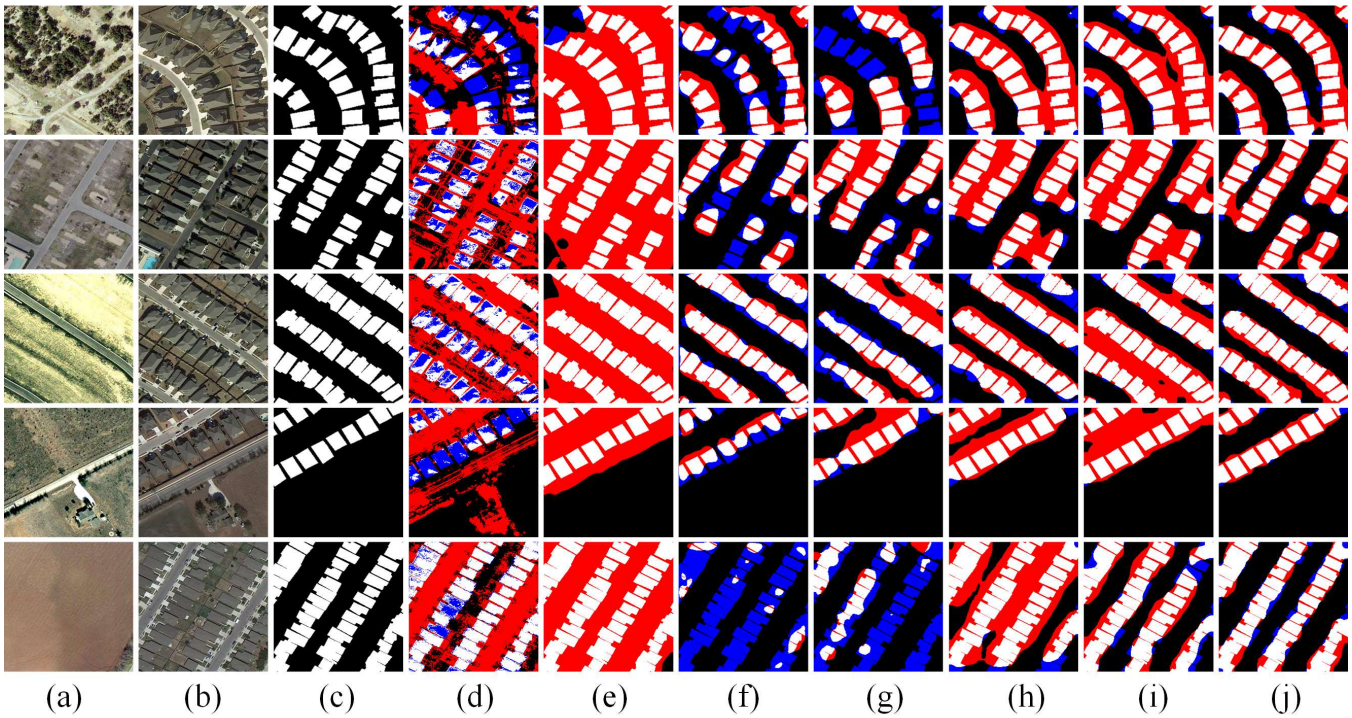



Fig. 8. Qualitative results of the proposed method compared with other methods on the LEVIR-CD dataset. (a) T1 image, (b) T2 image, (c) ground truth, (d) FCD-GAN [28], (e) CS-WSCDNet [41], (f) MS-Former [62], (g) TransWCD [63], (h) TransWCD-SA [63], (i) ACWCD [64], and (j) PDRNet. In the prediction results, white pixels represent the true positives, black pixels represent the true negatives, red pixels represent the false positives, and blue pixels represent the false negatives.

pseudo-label noise and enhances model robustness; yet, it also tends to suppress responses in low-confidence or ambiguous boundary regions, which may lead to a moderate decrease in recall. While this suppresses some uncertain or ambiguous changes and leads to a moderate reduction in recall, PDRNet achieves a much better balance between precision and recall. Consequently, it attains superior overall performance in comprehensive metrics such as $F1$-score and IoU, thereby improving the usability and credibility of the final change maps.

*2) Qualitative Evaluation:* The qualitative results of all methods on the three datasets are shown in Figs. 8–10. Among the compared methods, FCD-GAN can roughly localize most changed buildings, but the instability of adversarial training leads to noisy predictions. CS-WSCDNet suffers from a high false positive rate, as SAM may extend coarse CAM localization into adjacent unchanged areas with similar textures. TransWCD, relying only on deep features to generate CAM, loses spatial details and thus produces more false negatives. In contrast, MS-Former and TransWCD-SA reduce false positives, benefiting from bidirectional attention blocks and scene-adaptive gated predictors, respectively, which improve the prediction reliability.

In comparison, for the LEVIR-CD, where building changes are extremely dense, PDRNet benefits from WCEM to precisely localize change regions and delineate boundaries, thereby reducing both false positives and false negatives. In the case of WHU-CD, which contains diverse building morphologies, PDRNet leverages PAM to capture more complete change regions and employs the DRS to suppress the label noise, enabling accurate boundary delineation and fewer false detections. Within the GZ dataset, PDRNet still achieves the most reliable predictions, producing cleaner and structurally consistent change maps compared with other methods, as shown in Fig. 10. Overall, these visual comparisons highlight

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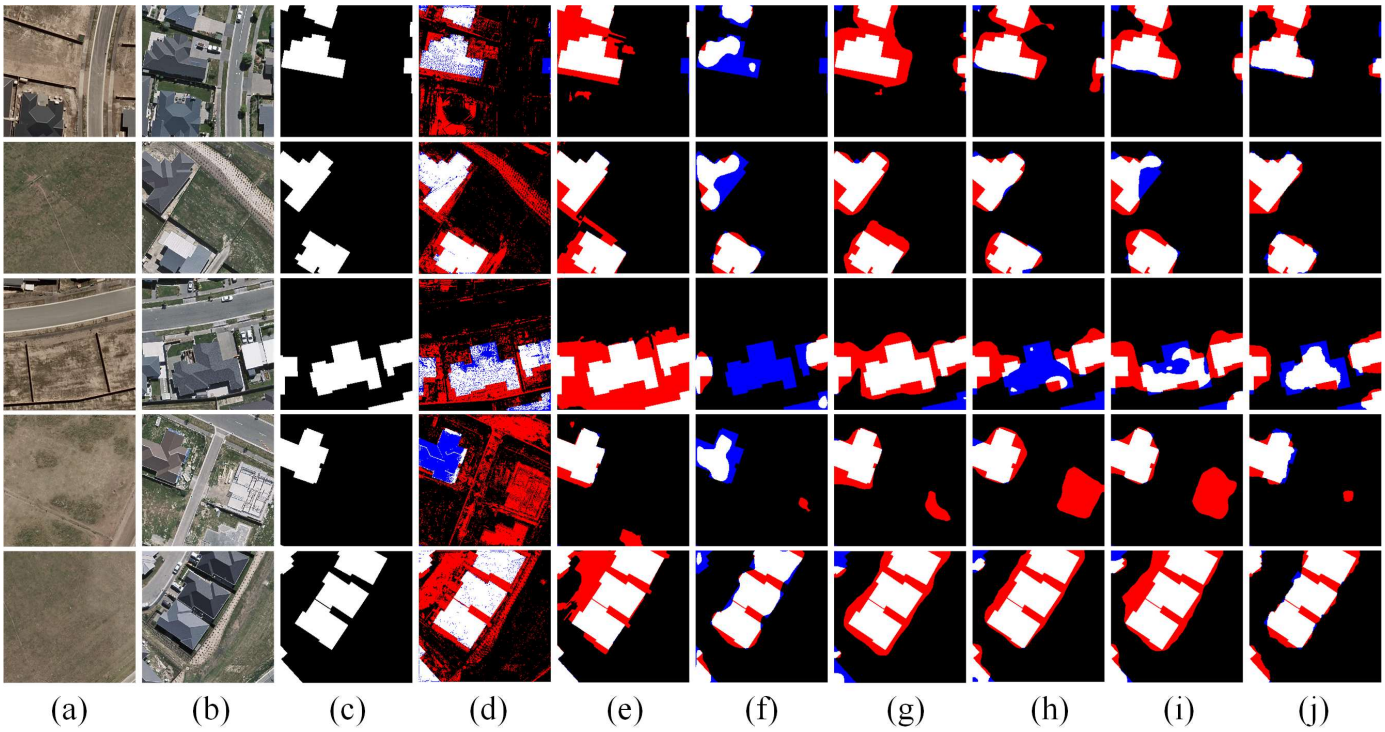



Fig. 9. Qualitative results of the proposed method compared with other methods on the WHU-CD dataset. (a) T1 image, (b) T2 image, (c) ground truth, (d) FCD-GAN [28], (e) CS-WSCDNet [41], (f) MS-Former [62], (g) TransWCD [63], (h) TransWCD-SA [63], (i) ACWCD [64], and (j) PDRNet. In the prediction results, white pixels represent true positives, black pixels represent true negatives, red pixels represent false positives, and blue pixels represent false negatives.

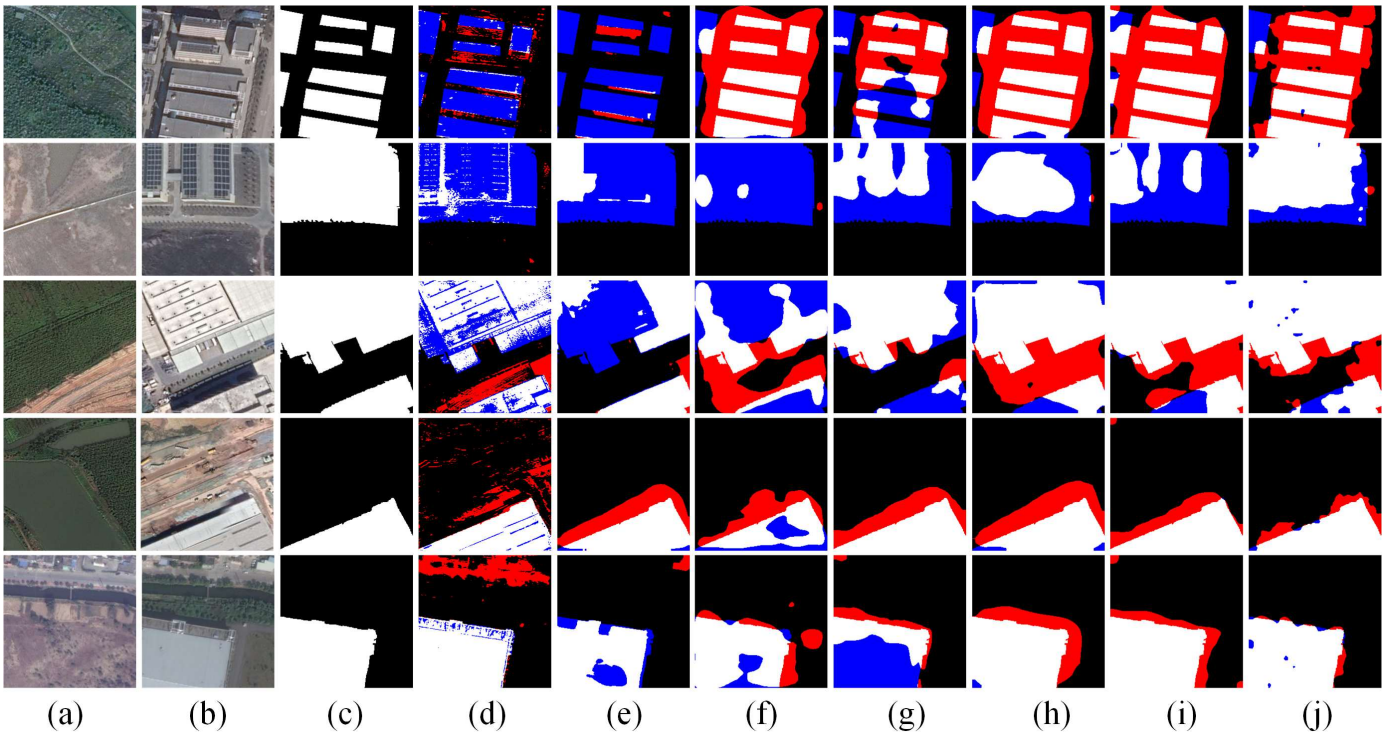
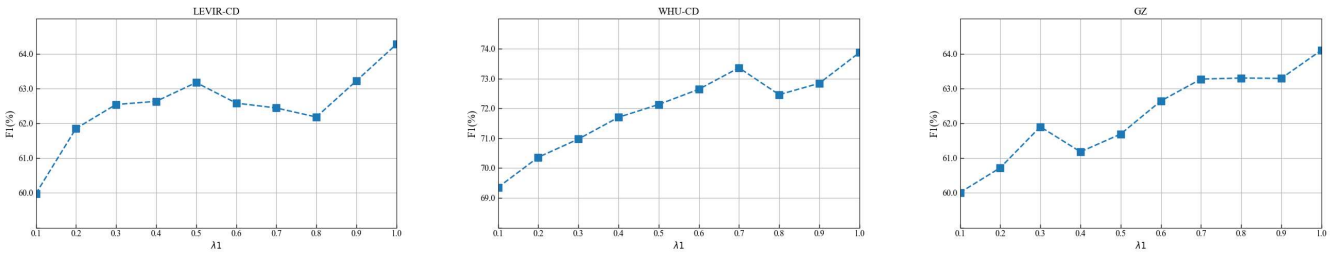



Fig. 10. Qualitative results of the proposed method compared with other methods on the GZ dataset. (a) T1 image, (b) T2 image, (c) ground truth, (d) FCD-GAN [28], (e) CS-WSCDNet [41], (f) MS-Former [62], (g) TransWCD [63], (h) TransWCD-SA [63], (i) ACWCD [64], and (j) PDRNet. In the prediction results, white pixels represent true positives, black pixels represent the true negatives, red pixels represent the false positives, and blue pixels represent the false negatives.

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(a) (b) (c)

Fig. 11. Effect of $\lambda_1$ on the quality of pseudo-labels. (a) LEVIR-CD, (b) WHU-CD, and (c) GZ.

TABLE V
IMPACT OF DIFFERENT $\tau$ VALUES ON THE QUALITY OF PSEUDO-LABELS

| $\tau$ | LEVIR-CD | | WHU-CD | | GZ | |
|---|---|---|---|---|---|---|
| | F1 | IoU | F1 | IoU | F1 | IoU |
| 0.5 | 61.82 | 44.73 | 67.61 | 51.07 | 59.07 | 41.92 |
| 0.6 | 62.53 | 45.48 | 72.83 | 57.28 | 63.02 | 46.00 |
| **0.7** | **64.27** | **47.36** | **73.87** | **58.57** | **64.10** | **47.17** |
| 0.8 | 62.84 | 45.81 | 71.11 | 55.17 | 62.96 | 45.95 |
| 0.9 | 62.58 | 45.54 | 73.17 | 57.69 | 62.38 | 45.33 |

TABLE VI
IMPACT OF DIFFERENT $\sigma$ VALUES

| $\sigma$ | LEVIR-CD | | WHU-CD | | GZ | |
|---|---|---|---|---|---|---|
| | F1 | IoU | F1 | IoU | F1 | IoU |
| 0.5 | 65.42 | 48.61 | 75.95 | 61.22 | 66.01 | 49.27 |
| 0.6 | 65.33 | 48.52 | 75.97 | 61.32 | 66.23 | 49.52 |
| 0.7 | 65.52 | 48.72 | 75.96 | 61.24 | 66.32 | 49.62 |
| 0.8 | 65.75 | 48.97 | 76.25 | 61.63 | 66.64 | 49.98 |
| **0.9** | **66.42** | **49.71** | **77.42** | **63.15** | **67.30** | **50.72** |

that PDRNet maintains superior capability across different scenarios by achieving both reliable localization and balanced precision–recall performance.

### *E. Parameter Analysis and Complexity Analysis*

In this section, the network parameters and computational complexity are analyzed. Among all hyperparameters, the momentum coefficient $\alpha$, the parameter $\beta$, and the standard optimizer settings (learning rate, momentum, and weight decay) adopt empirically validated values [31], [37]. In contrast, $\tau$, $\lambda_1$, $\lambda_2$, and $\sigma$ are critical factors that directly influence the quality of pseudo-label generation and the stability of training. Therefore, these four parameters are selected for further analysis.

*1) High-Confidence Pseudo-Label Threshold $\tau$:* The threshold $\tau$ controls the selection of high-confidence regions in the initial CAM. A smaller $\tau$ allows for the extraction of a broader range of high-confidence regions, which helps cover more potential targets. In contrast, a larger $\tau$ focuses on more precise regions; although spatially limited, these regions contain more reliable semantic information.

Table V presents the variations in pseudo label $F1$ scores and IoU corresponding to different values of $\tau$ across the three datasets. It can be observed that, on the three datasets, the quality of pseudo-labels improves initially and then declines as $\tau$ increases. This trend suggests that moderately increasing $\tau$ enhances the reliability of high-confidence regions, thereby facilitating the extraction of more stable class prototype representations. However, when $\tau$ becomes excessively large, the extracted regions become overly conservative, resulting in incomplete prototype representations and ultimately degrading overall performance. When the value of $\tau$ is set to 0.7, the best performance is achieved across all datasets. Therefore, $\tau$ is fixed at 0.7 in the experiments.

*2) Weight Factor $\lambda_1$ of $\mathcal{L}_{Sim}$:* The parameter $\lambda_1$ controls the strength of the similarity constraint among class prototypes. Fig. 11 illustrates the impact of different $\lambda_1$ values on the quality of pseudo-labels across the LEVIR-CD, WHU-CD, and GZ datasets, with $F1$ score used as the evaluation metric.

It can be observed that as $\lambda_1$ increases, the quality of pseudo-labels consistently improves. This is because, as $\lambda_1$ increases, the class prototypes become more discriminative. In PAM, these more discriminative prototypes reactivate the fused hierarchical features, thereby strengthening the alignment between semantic information and local details, and enhancing the completeness and semantic consistency of CAMs.

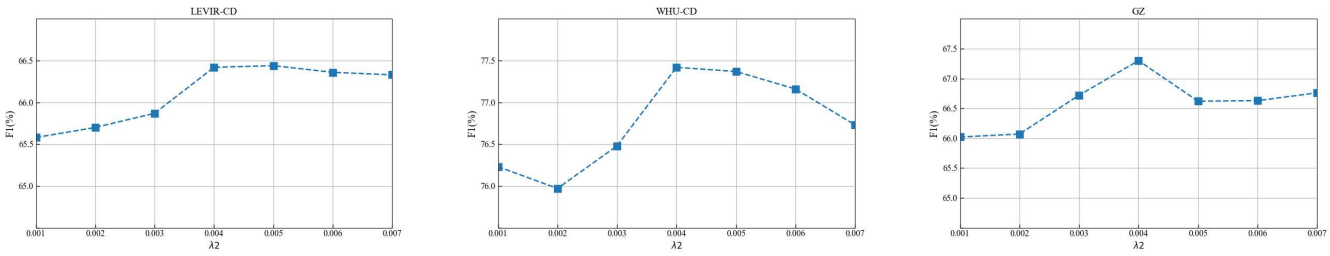
*3) Weight Factor $\lambda_2$ of $\mathcal{L}_{Con}$:* In this section, the analysis of the weight factor $\lambda$ is conducted. The impact of different values of $\lambda$ on PDRNet is explored across the three datasets. Fig. 12 presents the experimental results on three datasets, using $F1$ score as the evaluation metric.

As the hyperparameter $\lambda$ increases, the overall performance of PDRNet exhibits a rising-then-declining trend. This indicates that a properly tuned $\lambda$ facilitates effective guidance of the model via the consistency loss $\mathcal{L}_{\text{con}}$. It enhances alignment between predictions from strong and weak augmented views, enabling better exploitation of informative regions affected by label noise. However, when $\lambda$ exceeds an optimal threshold, the model may overly prioritize consistency across views. This can impair its ability to discriminate true change regions, ultimately resulting in performance degradation. According to the experimental results, the final $\lambda$ value is selected as 0.004

*4) Confidence Threshold $\sigma$:* The confidence threshold $\sigma$ influences the quality of the separation between clean regions and noisy regions within the pseudo-labels, thereby affecting the performance of model training. Table VI presents the variations in $F1$ score and IoU corresponding to different values of $\sigma$ on the three datasets.

It can be observed that as the value of $\sigma$ increases, the network performance improves accordingly. This is because a lower $\sigma$ introduces more interference from noisy labels, mistakenly treating low-confidence regions as clean, thereby

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(a) (b) (c)

Fig. 12. $F1$ of PDRNet with different a $\lambda_2$. (a) LEVIR-CD. (b) WHU-CD. (c) GZ.

TABLE VII
COMPARISON OF FLOPS AND PARAMETERS BETWEEN THE PROPOSED METHOD AND SIX COMPARISON METHODS

| Method | Params (M) | FLOPs (G) |
|---|---|---|
| FCD-GAN [28] | 43.35 | 161.78 |
| CS-WSCDNet [41] | 29.52 | 47.86 |
| MS-Former [62] | 15.20 | 22.14 |
| TransWCD [63] | 13.68 | 3.86 |
| TransWCD-SA [63] | 17.87 | 8.92 |
| ACWCD [64] | 18.07 | 8.04 |
| PDRNet | 46.18 | 7.52 |

providing incorrect supervisory signals. In contrast, a higher $\sigma$ helps filtering out uncertain noisy regions, allowing the training process to focus more on high-confidence, reliable clean regions, thus enhancing overall performance. According to the experimental results, the final value of $\sigma$ is set to 0.9.

*5) Complexity Analysis:* To assess model scalability and computational efficiency, Table VII presents a comparison of FLOPs and parameter counts between PDRNet and existing approaches. Among them, CS-WSCDNet and PDRNet present the relevant parameters and FLOPs of the CD network. Although PDRNet has a higher number of parameters, it maintains a relatively low computational cost with only 7.52 GFLOPs. This is primarily attributed to the high-dimensional prototype storage and the additional convolutional operations introduced by the WCEM in PDRNet, which can increase the overall parameter count. Nevertheless, these components rely on computationally lightweight operations, such as similarity measurement and linear wavelet transformation. Furthermore, DRS only applies data augmentation to input samples and utilizes simple thresholding and confidence estimation, contributing minimal inference overhead. Compared with models such as FCD-GAN and CS-WSCDNet, PDRNet achieves a good balance between accuracy and efficiency. It reduces computational overhead while maintaining high detection performance. These results indicate that PDRNet offers strong computational efficiency, making it suitable for large-scale data processing and real-time applications.

## IV. CONCLUSION

In this article, a PDRNet is proposed for VHR remote sensing images WSCD. Initially, PDRNet employs a PAM to capture globally stable prototypes and reactivate fused hierarchical features. By leveraging semantic matching, PAM identifies latent change regions that were previously insufficiently activated, substantially enhancing the completeness of CAM. Subsequently, to alleviate the detrimental effects of pseudo-label noise, DRS partitions pseudo-labels into clean and noisy subsets based on predictions from strong and weak augmentations of the same sample. This strategy enables the network to leverage reliable supervision signals, effectively mitigating overfitting to label noise. Furthermore, WCEM decomposes bitemporal features into distinct frequency components, capturing structural semantic changes in the low-frequency while enhancing texture detail changes in the high frequency, which significantly improves the accuracy of CD. Conclusively, experimental results on three publicly available datasets show that PDRNet achieves superior performance compared with six state-of-the-art weakly supervised CD methods. In future work, we will further explore methods to improve recall while maintaining high precision, thereby enabling more complete and accurate identification of change regions.

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